Ferroelectric Vortex Memory Platform

Complete Engineering Specification

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11 September 2026

Document Type: Engineering Design — Core Specification
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Technology Patent Status: **PUBLIC RELEASE — NO PATENT SOUGHT**
Advanced Optimization: *Available later in this document*
Repository: Mega — Proton Drive — Zenodo — GitHub — OSF — Academia — X

This document provides the complete engineering specification for a non-volatile memory device based on ferroelectric vortex cells. All cell geometries, material compositions, fabrication steps, circuit designs, operating procedures, qualification tests, and acceptance criteria are provided explicitly. Every parameter required to build, test, and operate the baseline device is included. The design is released as a public, patent-free engineering specification. An advanced optimization configuration is offered separately and is described at the end of this document.

This is a 6DUFT inspired technology.

Abstract

This document is the complete engineering specification for a non-volatile memory device based on ferroelectric vortex cells.

Each cell stores information in the topological winding number of a ferroelectric vortex pinned at a central TiN pillar. The vortex core is conductive, enabling resistance-based readout. The device provides non-volatile storage with low read latency, very low write energy, and very low power consumption.

The baseline device provides:

- **Density:** 1.95 GB/cm^2
- **Capacity:** 1.95 GB on a 100 mm^2 die
- **Read latency:** 3.80 ns
- **Write latency:** 21.1 ns
- **Write energy:** 595 fJ per bit
- **Endurance:** 2.74×10^{10} cycles
- **Retention:** >10 years at 85°C
- **Power:** 97.2 mW on a 100 mm^2 die
- **Die cost:** $5.36
- **Chip cost:** $7.36
- **Cost per GB:** $2.75 (die-cost basis)

The device uses standard 28 nm CMOS with hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) atomic layer deposition. No exotic materials are required. The complete fabrication flow has 18 steps and 10 masks.

The document covers cell architecture, material specifications, the complete fabrication flow, circuit design, operating principles, performance specifications, verification and qualification, failure modes and mitigations, packaging, and an advanced optimization configuration available separately.

The design is released openly. No patent is sought.

This is a 6DUFT inspired technology.

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Part I
Introduction and Overview

1 Purpose and Scope

This document is the complete engineering specification for a non-volatile memory device based on ferroelectric vortex cells.

It provides every parameter, process step, circuit specification, and procedure required to build, test, and operate the baseline device.

The document is written for:

- Foundry process engineers who will fabricate the device.
- Circuit designers who will integrate the device.
- Test engineers who will qualify the device.
- System engineers who will deploy the device.

No prior knowledge of any theoretical framework is required. All necessary engineering information is contained in this document.

2 What the Device Is

The device is a non-volatile memory that stores information in the topological winding number of a ferroelectric vortex.

Each memory cell consists of:

- A ferroelectric nanodisk made of hafnium zirconium oxide (HZO).
- A central TiN pinning pillar.
- A bottom electrode (TiN).
- A top electrode (TiN).
- A select transistor (FinFET).

The polarization field in the ferroelectric disk forms a vortex around the pinning pillar. The vortex has an integer winding number n that describes how many times the polarization rotates around the pillar.

The vortex core is a narrow region where the polarization is suppressed. The core is conductive because the polarization suppression creates a deep potential well with high carrier density.

The winding number determines the core geometry, which determines the electrical resistance between the top and bottom electrodes. Reading the resistance reveals the stored winding number.

For binary operation:

- $n = 0$: uniform polarization, no vortex, resistance $> 10^{12}$ Ω.
- $n = 1$: single vortex, resistance in the kΩ range.

The ON/OFF ratio exceeds 10^9, making binary discrimination straightforward.

3 What the Device Is Not

The device is not:

- A DRAM replacement. DRAM has higher endurance (10^{16} cycles vs. 2.74×10^{10} cycles). This device is for applications where non-volatility, low power, and high density matter more than unlimited endurance.
- A NAND replacement at the baseline tier. NAND has lower cost per bit at the baseline tier. The device becomes cheaper than NAND at the advanced optimization configuration.
- A compute device. The device is a storage device. It does not perform matrix-vector multiplication.
- A general-purpose memory. The write-verify procedure adds latency to writes. The device is optimized for read-heavy workloads.

4 Key Advantages

- **Non-volatile:** Data is retained without power. 10+ year retention at 85°C.
- **High density:** 1.95 GB/cm^2 at the baseline tier, which is 15.6× denser than DRAM.
- **Fast read:** 3.80 ns read latency, which is 13,158× faster than NAND flash.
- **Low write energy:** 595 fJ per bit, which is 16.8× lower than DRAM.
- **Very low power:** 97.2 mW for a 100 mm^2 die, which is 31–51× lower than DRAM.
- **CMOS compatible:** Standard 28 nm CMOS with HZO atomic layer deposition. No exotic materials.

5 Key Limitations

- **Lower endurance than DRAM:** 2.74×10^{10} cycles vs. 10^{16} cycles. Wear leveling is required for write-intensive applications.
- **Write-verify required:** The write operation requires a verify step to ensure reliable switching, increasing the write latency.
- **Baseline cost:** The baseline cost per GB is higher than NAND flash. The cost advantage appears at the advanced optimization configuration.

6 Device Positioning

The baseline device is positioned for:

- Embedded memory
- Low-volume production
- Educational and research use
- Applications where the HZO process is already established

An advanced optimization configuration with higher density, lower write energy, and lower cost per GB is available separately and is described at the end of this document.

Part II
Device Architecture

7 Cell Structure

7.1 Layer Stack

Each memory cell is a vertical stack of the following layers, from bottom to top:

Layer	Material	Thickness	Function
Bottom electrode	TiN	50 nm	Electrical contact
Etch stop	Al_2O_3	2 nm	Protects bottom electrode during pillar etch
Pillar	TiN	1.96 nm	Vortex pinning center
Ferroelectric	$Hf_{0.5}Zr_{0.5}O_2$	14.83 nm	Storage medium
Top electrode	TiN	50 nm	Electrical contact

7.2 Top-Down Geometry

The top-down geometry of the cell is:

Parameter	Value
Cell size	80 nm × 80 nm
Disk diameter	40 nm
Pillar diameter	20 nm
Annular width (pillar to disk edge)	10 nm

The ferroelectric disk is circular, centered on the pillar. The annular region between the pillar and the disk edge is where the vortex forms.

7.3 Select Transistor

Each cell is connected to a select transistor.

Parameter	Value
Type	FinFET
Fins	3
Gate length	28 nm
Gate oxide	High-κ (HfO_2, EOT 3 nm)
Source/drain	$CoSi_2$
Threshold voltage	0.3 V
On-current	299 μA

8 Array Organization

8.1 Bank Structure

Parameter	Value
Number of banks	16
Capacity per bank (100 mm^2)	122 MB
Bank address bits	4

Up to 4 banks can be active simultaneously for parallel reads and writes.

8.2 Row and Column Organization

Parameter	Value
Rows per bank	1024
Columns per bank	1024
Word lines per bank	1024
Bit lines per bank	1024

9 Readout Architecture

Each bit line is connected to a comparator that discriminates between the two resistance states.

Parameter	Value
Comparator type	Current-mode with auto-zeroing
Decision time	199.5 ps
Supply voltage	1.0 V
Number of comparators	1024 (one per bit line)

The comparator includes an auto-zeroing circuit that cancels offset. Auto-zeroing is scheduled during idle cycles and does not affect read latency.

10 Write Architecture

Each word line is connected to a write driver that applies the write voltage.

Parameter	Value
Write driver type	H-bridge
Write voltage range	-3.2 V to $+3.2$ V
Rise time	300 ps
Fall time	300 ps
Transistor type	5 V I/O

A negative charge pump generates the erase voltage.

Parameter	Value
Charge pump type	Dickson 4-stage
Input	$+1$ V
Output	-3.2 V
Efficiency	80%
Power	302 μW

Part III
Cell Geometry and Materials

11 Cell Dimensions

Parameter	Value
Cell size	80 nm × 80 nm
Disk diameter	40 nm
Disk radius	20 nm
Pillar diameter	20 nm
Pillar radius	10 nm
Pillar height	1.96 nm
Annular width	10 nm
Film thickness	14.83 nm

12 Annular Area

The annular area for $n = 1$ is:

$$A_1 = \pi[(r_{\text{pillar}} + r_{\text{core}})^2 - r_{\text{pillar}}^2] \tag{1}$$

where r_{core} is the vortex core radius. With $r_{\text{pillar}} = 10$ nm and a vortex core radius of 2.31 nm:

$$A_1 = \pi[(10 \text{ nm} + 2.31 \text{ nm})^2 - (10 \text{ nm})^2] = 161.9 \text{ nm}^2 \tag{2}$$

13 Material Properties

13.1 Hafnium Zirconium Oxide (HZO)

Property	Value
Composition	$Hf_{0.5}Zr_{0.5}O_2$
Film thickness	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 μC/cm^2
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m
Film uniformity	±1% within wafer

13.2 Core Properties

Property	Value
Core radius	2.31 nm
Core conductivity	7.79×10^3 S/m

13.3 Electrode Materials

Electrode	Material	Thickness
Bottom	TiN	50 nm
Top	TiN	50 nm
Pillar	TiN	1.96 nm
Etch stop	Al_2O_3	2 nm

14 Cell Electrical Characteristics

Parameter	Value
Resistance ($n = 1$)	11,762 Ω
Resistance ($n = 0$)	$> 10^{12}$ Ω
ON/OFF ratio	$> 10^9$
Capacitance	23.38 aF
Settling time	0.275 ps
Read voltage	9.886 mV
Read current ($n = 1$)	840.5 nA

Part IV
Fabrication Flow

15 Overview

The fabrication flow has 18 steps and uses 10 masks. The flow uses standard 28 nm CMOS front-end processing with HZO atomic layer deposition for the ferroelectric layer.

16 Step-by-Step Process Flow

16.1 Step 1: Substrate Preparation

Parameter	Value
Wafer size	300 mm
Substrate	Silicon (100)
Thermal SiO_2	100 nm ± 2 nm
Cleaning	RCA clean before oxide
Surface preparation	HF dip, DI rinse, N_2 dry

16.2 Step 2: FinFET Fabrication

Standard 28 nm CMOS front-end.

Parameter	Value
Gate length	28 nm
Fins	3-fin
Gate oxide	High-κ (HfO_2, EOT 3 nm)
Source/drain	$CoSi_2$
Threshold voltage	0.3 V

16.3 Step 3: Bottom Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300°C
Precursors	$TiCl_4$ + NH_3
Growth rate	0.05 nm/cycle
Cycles	1000

16.4 Step 4: Etch Stop Deposition

Parameter	Value
Material	Al_2O_3
Thickness	2 nm ± 0.2 nm
Method	ALD
Temperature	300°C
Precursors	$Al(CH_3)_3$ + H_2O

16.5 Step 5: Pillar Layer Deposition

Parameter	Value
Material	TiN
Thickness	1.96 nm ± 0.1 nm
Method	ALD
Temperature	300°C

16.6 Step 6: Pillar Patterning

Parameter	Value
Diameter	20 nm
Method	Immersion lithography (193 nm)
Alignment tolerance	±2 nm

16.7 Step 7: Pillar Etch

Parameter	Value
Chemistry	Cl_2/Ar
Method	RIE
Etch rate	∼1 nm/s
Etch stop	Al_2O_3 layer
Overetch	10%
Selectivity to Al_2O_3	>10:1

16.8 Step 8: Etch Stop Removal

Parameter	Value
Etchant	Dilute HF (0.1%)
Al_2O_3 etch rate	0.5 nm/s
TiN etch rate	<0.01 nm/s
Etch time	4 s (removes 2 nm, 2× overetch)

After this step, the Al_2O_3 remains only under the TiN pillar.

16.9 Step 8b: Post-Etch Cleaning

Step	Description
Rinse	Deionized water, 30 s
Dry	N_2 blow dry, 30 s
Plasma clean	Ar plasma, 30 s at 50 W

16.10 Step 9: HZO Deposition

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$
Thickness	14.83 nm
Method	ALD
Temperature	250°C
Supercycles	82
Deposition time	17 min

Process note: The film must be annealed within 1 hour of deposition to prevent moisture absorption.

16.11 Step 10: RTA Annealing

Parameter	Value
Method	Rapid Thermal Annealing
Temperature	450°C
Duration	20 s
Atmosphere	N_2
Ramp rate	50°C/s

16.12 Step 11: Top Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300°C

16.13 Step 12: Top Electrode Patterning

Parameter	Value
Pattern	Circle
Diameter	40 nm
Alignment	Centered on pillar
Tolerance	±2 nm

16.14 Step 13: Disk Etch

Parameter	Value
Stack	TiN (50 nm) + HZO (14.83 nm) + TiN (50 nm)
Chemistry (TiN)	Cl_2/Ar
Chemistry (HZO)	BCl_3/Ar
Etch stop	SiO_2 buried oxide
Selectivity to SiO_2	>3:1

Multi-step etch:

1. TiN etch (50 nm): Cl_2/Ar, 50 s.
2. HZO etch (14.83 nm): BCl_3/Ar, 15 s.
3. TiN etch (50 nm): Cl_2/Ar, 50 s.

Endpoint detection: optical emission spectroscopy. Ti line at 363.5 nm, Hf line at 313.5 nm.

16.15 Step 14: Post-Etch Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	O_2
Purpose	Repair sidewall damage

16.16 Step 15: Passivation

Parameter	Value
Material	SiN (50 nm) + SiO_2 (50 nm)
Method	PECVD
Temperature	350°C

16.17 Step 16: Contact Via Formation

Parameter	Value
Via diameter	50 nm
Fill material	Tungsten (CVD)
Resistivity	5.6×10^{-8} Ω·m

16.18 Step 17: Cu Interconnect

Parameter	Value
Thickness	1000 nm
Barrier	Ta/TaN (10 nm)
Seed	Cu (50 nm PVD)
Fill	Cu electroplating
CMP	After fill
Sheet resistance	0.017 Ω/□ (for 1000 nm Cu)

Increasing the metal thickness reduces the sheet resistance proportionally and lowers the RC delay of the interconnect.

16.19 Step 18: Final Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	Forming gas (5% H_2 in N_2)
Purpose	Contact sintering, interface passivation

17 Mask Set

Mask	Purpose	CD	Type
M1	FinFET gate	28 nm	Dark field
M2	Bottom electrode	100 nm	Dark field
M3	Pillar pattern	20 nm	Dark field
M4	Top electrode	40 nm	Dark field
M5	Disk pattern	40 nm	Clear field
M6	Contact vias	50 nm	Dark field
M7	Cu interconnect	1000 nm	Dark field
M8	Bond pad opening	50 μm	Clear field
M9	Passivation via	50 nm	Dark field
M10	Scribe line test structures	—	—

Total masks: 10.

18 Thermal Budget

Step	Temperature	Limit
FinFET	∼1000°C	$CoSi_2$ stable to 700°C
$CoSi_2$ formation	∼450°C	
Bottom electrode	300°C	TiN stable to 700°C
RTA anneal	450°C	HZO phase 400–600°C
Post-etch anneal	400°C	
Final anneal	400°C	Cu stable to 400°C

All temperatures are within limits.

19 Yield Analysis

19.1 Defect Density

Defect Type	Density
HZO pinholes	<0.1 cm^{-2}
TiN pillar defects	<0.01 cm^{-2}
Disk etch defects	<0.1 cm^{-2}
Passivation defects	<1 cm^{-2}
Particle contamination	<1 cm^{-2}
Total (mature)	**0.5 cm^{-2}**

19.2 Yield by Die Size

For a 100 mm^2 die:

$$Y = e^{-0.5 \times 1.0} = 60.7\% \tag{3}$$

Good dies per 300 mm wafer: 615 total, 373 good.

20 Cost Analysis

Parameter	Value
Wafer cost	\$2,000
Good dies per wafer	373
Die cost	\$5.36
Package cost	\$1.50
Test cost	\$0.50
Total chip cost	\$7.36
Capacity (100 mm^2)	1.95 GB
Cost per GB	\$2.75

Part V
Circuit Design

21 Comparator

21.1 Circuit Description

The comparator discriminates between the two resistance states by comparing the cell current to a reference current.

Comparator Topology
Input current → Current mirror → Integration node → Latch → Output

The current-mirror input stage converts the input current to a voltage on the integration node. The regenerative latch then amplifies the voltage difference to a full-swing digital output.

21.2 Auto-Zeroing

The comparator offset is cancelled by auto-zeroing:

1. Phase 1: Input shorted, measure offset, store on capacitor.
2. Phase 2: Input connected, measure signal, subtract offset.

Auto-zeroing adds approximately 1 ns to the read time but eliminates offset errors. It is scheduled during idle cycles and does not impact the read latency.

21.3 Specification

Parameter	Value
Type	Current-mode comparator
Topology	Double-tail latch with pre-amplification
Decision time	199.5 ps
Supply voltage	1.0 V
Power per comparator	0.840 μW
Offset (after auto-zero)	$<$10 μV
Number	1024 per bank

22 Word Line Driver

Parameter	Value
Type	CMOS buffer
Output voltage	0–1 V
Rise time	100 ps
Drive capability	299 μA
Power	50 μW per driver
Number	1024 per bank

23 Write Driver

Parameter	Value
Topology	H-bridge
Voltage range	−3.2 V to +3.2 V
Rise time	300 ps
Fall time	300 ps
Current capability	10 μA
Transistor type	5 V I/O

24 Charge Pump

Parameter	Value
Topology	Dickson 4-stage
Input	+1 V
Output	−3.2 V
Efficiency	80%
Power	302 μW
Area	50×100 μm

25 Voltage Reference

Parameter	Value
Type	Bandgap reference
Accuracy	±1%
Temperature coefficient	<50 ppm/°C
Power	100 μW

26 ECC

Parameter	Value
Code type	SEC (Hamming)
Data word	64 bits
Check bits	7 bits
Code rate	0.90
Overhead	11.1%
Correctable errors	1 per 64 bits
Power	10 mW

The ECC encoding and decoding are pipelined and do not add to the read or write latency.

27 Bad Block Management

1. At test time, identify bad blocks.
2. Store bad block list in reserved area (block 0).
3. During operation, skip bad blocks.

4. Periodically scan for new bad blocks.

Bad block table size: 10% of total block count.

28 Wear Leveling

1. Controller maintains write count for each block.
2. New writes go to the least-used block.
3. Old blocks are marked for recycling.

Write count storage: 32 bits per block. For 15,625 blocks, the write count table is 62.5 KB.

29 Power Management

State	Description	Wake Time
P0	Full operation	—
P1	Reduced clock (50%)	10 ns
P2	Idle (clock gated)	1 ns
P3	Sleep (power removed)	10 ns

Data is non-volatile. Power-off is instant. Power-on restores the device with no state loss.

30 Clock Generation

Parameter	Value
Reference	100 MHz
PLL	Charge-pump PLL
Output frequency	100 MHz to 1 GHz
Jitter	$<$1 ps RMS
Lock time	$<$10 μs
Power	10 mW

31 Temperature Sensor

Parameter	Value
Type	Bandgap-based
Accuracy	$\pm$1°C
Resolution	0.25°C
Power	1 μW
Number	4 per die

32 Complete Power Summary

Component	Power
Comparators	0.860 mW
Control	20 mW
I/O	50 mW
Clock	10 mW
ECC	10 mW
Charge pump	0.302 mW
Leakage	6 mW
Total	**97.2 mW**

Part VI
Operating Principles

33 Write Operation

33.1 Write Procedure

For each cell:

1. Decode bank address (1 ns).
2. Decode row address (0.5 ns).
3. Activate word line (0.1 ns).
4. Apply write voltage attempt 1 (10 ns).
5. Verify (read back, 2 ns).
6. If failed, apply attempt 2 (10 ns).
7. Verify (2 ns).
8. If failed, apply attempt 3 (10 ns).
9. Verify (2 ns).
10. If failed, apply attempt 4 (10 ns).
11. Verify (2 ns).

33.2 Write-Verify Voltage Levels

Attempt	Field	Voltage	Per-Level	Cumulative
1	1.5 MV/cm	2.2245 V	0.500	0.500
2	1.7 MV/cm	2.5211 V	0.375	0.875
3	1.9 MV/cm	2.8177 V	0.109	0.984
4	2.1 MV/cm	3.1143 V	0.0146	0.999

The remaining 0.1% of cells that fail all four attempts are handled by ECC and bad block management.

33.3 Average Write Time

$$t_{\text{avg}} = 1.5 + 10 + 2 + 0.375 \times 12 + 0.109 \times 24 + 0.0146 \times 36 = 21.1 \text{ ns} \tag{4}$$

33.4 Write Energy

Component	Energy per bit
Switching	548.2 fJ
Capacitive	0.116 fJ
Vortex	0.523 fJ
Weighted total	**595 fJ**

34 Read Operation

34.1 Read Procedure

1. Decode bank address (1.0 ns).
2. Decode row address (0.5 ns).
3. Activate word line (0.1 ns).
4. Cell settling (0.275 ps).
5. Comparator decision (199.5 ps).
6. Data output (2.0 ns).

34.2 Read Latency Breakdown

Component	Time
Bank decode	1.000 ns
Row decode	0.500 ns
Word line	0.100 ns
Settling	0.0003 ns
Comparator	0.1995 ns
Data output	2.000 ns
Total	**3.80 ns**

34.3 Read Disturb Analysis

The read field is 0.00667 MV/cm at 85°C. The coercive field at 85°C is 1.18 MV/cm. The margin is 177×. No read disturb occurs over the device lifetime.

35 Erase Operation

The erase operation writes $n = 0$ to the cell. It uses the same write-verify procedure with the opposite polarity.

Parameter	Value
Erase voltage range	-2.2245 to -3.1143 V
Erase energy	Same as write energy

36 Temperature Compensation

The coercive field decreases with temperature:

$$E_c(T) = E_c(0) \times (1 - T/579\ \text{K}) \tag{5}$$

The write voltage is adjusted by approximately 19% across the operating range.

Parameter	Value
Range	0–85°C
Resolution	1°C
Points	86
Voltage rails	4
Total entries	688
Size	688 bytes

37 Periodic Recalibration

Reference cells (1% of the array) are programmed to known states during initial calibration.

Recalibration procedure:

1. Read all reference cells.
2. Compare to stored values.
3. Update comparator thresholds.
4. Update write voltage lookup table.

Frequency: Hourly during normal operation. Time: 1 ms (background operation).

38 Power-On Sequence

1. Apply 1 V logic supply.
2. Wait for power-on reset (1 μs).
3. Apply write voltage supply (from charge pump).
4. Load calibration data from non-volatile memory.
5. Begin normal operation.

Total boot time: $<$10 μs.

39 Power-Down Sequence

1. Complete pending operations.
2. Remove write voltage supply.
3. Remove logic supply.

Data is non-volatile. Power-off is instant. No data loss.

40 Error Handling

40.1 Correctable Errors

1. Correct the error using ECC.
2. Log the event.
3. Continue operation.

40.2 Uncorrectable Errors

1. Retry the operation.
2. If retry fails, mark block as bad.
3. Use a spare block.
4. If no spares remain, report to host.

41 Wake-Up Procedure

Production test performs the wake-up:

1. Apply 10^4 write cycles to all cells.
2. Measure resistance after wake-up.
3. Store calibration values.
4. Begin normal operation.

Wake-up time: $10^4 \times 21.1$ ns $= 211$ μs.

Part VII
Performance Specifications

42 Summary of Baseline Performance

Parameter	Value
Density	1.95 GB/cm^2
Capacity (100 mm^2 die)	1.95 GB

Parameter	Value
Read latency	3.80 ns
Write latency (average)	21.1 ns
Write energy	595 fJ per bit
Endurance	2.74×10^{10} cycles
Retention at 85°C	>10 years
Power (100 mm^2 die)	97.2 mW
Die cost	$5.36
Chip cost	$7.36
Cost per GB	$2.75
Operating temperature	0–85°C
Supply voltage	1.0 V logic, −3.2 V write/erase

43 Comparison with Existing Technology

43.1 Comparison with DRAM

Parameter	DRAM	This Device
Density	0.125 GB/cm^2	1.95 GB/cm^2
Read latency	50 ns	3.80 ns
Write energy	10 pJ	595 fJ
Write latency	50 ns	21.1 ns
Endurance	10^{16} cycles	2.74×10^{10} cycles
Non-volatile	No	Yes
Power (100 mm^2)	3–5 W	97.2 mW
Cost per GB	$5–10	$2.75

43.2 Comparison with NAND Flash

Parameter	NAND	This Device
Density	0.375 GB/cm^2	1.95 GB/cm^2
Read latency	50 μs	3.80 ns
Write energy	1 nJ	595 fJ
Write latency	100 μs	21.1 ns
Endurance	10^4–10^5 cycles	2.74×10^{10} cycles
Non-volatile	Yes	Yes
Cost per GB	$0.50–1	$2.75

44 Honest Assessment

44.1 Strengths

- 15.6× denser than DRAM.
- 13,158× faster read than NAND flash.
- 16.8× lower write energy than DRAM.

- 31–51× lower power than DRAM.
- Non-volatile with 10+ year retention.
- CMOS compatible with standard HZO ALD.
- No exotic materials.

44.2 Weaknesses

- Endurance is lower than DRAM. Wear leveling is required for write-intensive applications.
- Write-verify procedure adds latency to writes.
- Baseline cost per GB is higher than NAND flash.

Part VIII
Verification and Qualification

45 Overview

The verification plan is designed to catch problems at the earliest possible stage, minimizing wasted effort and fabrication runs.

The plan follows the three-gate structure:

1. **Gate 1:** The ferroelectric phase is confirmed immediately after annealing.
2. **Gate 2:** The vortex cell switches correctly at the single-cell level.
3. **Gate 3:** The device works at scale in an array.

46 Gate 1: Ferroelectric Phase Confirmation

46.1 When to Measure

After Step 10 (RTA Annealing), before any further patterning.

46.2 What to Measure

A P-V hysteresis loop on a simple TiN/HZO/TiN capacitor.

46.3 Test Structure

A 50 μm × 50 μm capacitor fabricated on the same wafer, in the scribe line or a dedicated test area.

46.4 Equipment

Radiant Precision LC ferroelectric tester, or equivalent.

46.5 Procedure

1. Contact the top and bottom electrodes with probe tips.
2. Apply a voltage sweep from -5 V to $+5$ V and back.
3. Measure the polarization as a function of voltage.
4. Plot the P-V loop.

46.6 Pass Criteria

Measurement	Target	Pass
Hysteresis loop	Open loop	Clearly visible
Remanent polarization	>10 μC/cm^2	Measured above threshold
Coercive field	1–2 MV/cm	Within range
Film thickness	14.83 nm $\pm$ 1%	Ellipsometry

46.7 What Passing Gate 1 Tells You

The ferroelectric has crystallized in the orthorhombic phase. The material foundation is solid. The annealing conditions are correct. The ALD deposition worked.

46.8 If Gate 1 Fails

Symptom	Likely Cause	Fix
No hysteresis	Not crystallized	Increase anneal to 500°C
Low P_s	Low phase fraction	Increase anneal time, check wake-up
Wrong coercive field	Wrong film thickness	Check ALD cycle count
Leaky capacitor	Pinholes	Check contamination, ALD QC

47 Gate 2: Single Cell Verification

47.1 When to Measure

After Step 13 (Disk Etch) and Step 14 (Post-Etch Anneal), after the top electrode is patterned.

47.2 What to Measure

Write/read operation on a single isolated cell.

47.3 Test Structure

A single cell with the full stack: TiN bottom, TiN pillar, HZO, TiN top. Isolated from other cells.

47.4 Equipment

Semiconductor parameter analyzer (Keysight B1500A or equivalent), probe station with micromanipulators.

47.5 Procedure

1. Probe the top electrode and the bottom electrode.
2. Apply the write voltage (2.2245 V) for 10 ns (write $n = 1$).
3. Apply the read voltage (9.886 mV) and measure the current.
4. Apply the erase voltage (-2.2245 V) for 10 ns (write $n = 0$).
5. Apply the read voltage and measure the current.
6. Repeat 10 times to verify reproducibility.

47.6 Expected Results

Parameter	Value
$n = 1$ read current	840.5 nA
$n = 0$ read current	<1 pA
ON/OFF ratio	$> 10^9$
Reproducibility	10/10 cycles

47.7 Pass Criteria

Measurement	Target	Pass
$n = 1$ read current	Within 50% of expected	Measured in range
$n = 0$ read current	<1 pA	Below threshold
ON/OFF ratio	$> 10^9$	Measured above threshold
Reproducibility	10/10 cycles	All succeed

47.8 If Gate 2 Fails

Symptom	Likely Cause	Fix
$n = 1$ current too low	Resistance too high	Check pillar diameter, film thickness
$n = 0$ current too high	Leakage path	Check disk etch completeness
No switching	No vortex	Check pillar height, annular width
Inconsistent switching	Alignment error	Check pillar-to-disk alignment

48 Gate 3: Small Array Verification

48.1 When to Measure

After full fabrication (Step 18), before dicing and packaging.

48.2 What to Measure

Array-level yield and variation on a 64×64 test array.

48.3 Test Structure

A 64×64 array (4,096 cells) with full addressing and readout.

48.4 Equipment

Automated wafer prober with test program.

48.5 Procedure

1. Write all 4,096 cells to $n = 1$.
2. Read all cells and measure current.
3. Write all cells to $n = 0$.
4. Read all cells and measure current.
5. Calculate yield and cell-to-cell variation.

48.6 Pass Criteria

Measurement	Target	Pass
Array yield	>80%	Measured above threshold
Cell variation	<20% (1σ)	Measured below threshold
Write/read success	100%	No systematic failures

48.7 If Gate 3 Fails

Symptom	Likely Cause	Fix
Low yield	Random defects	Cleanroom, process cleanliness
High variation	ALD non-uniformity	Check ALD tool, recipe
Systematic failure	Design error	Review layout, DRC
Edge failures	Etch non-uniformity	Check etch recipe, overetch

49 Additional Intermediate Tests

Step	Measurement	Target	Method
After HZO dep	Film thickness	±1%	Ellipsometry
After HZO dep	Composition	Hf:Zr 50:50 ±2%	XPS
After anneal	Phase fraction	50–70% orthorhombic	GI-XRD
After pillar etch	Pillar diameter	±10%	SEM
After disk etch	Disk diameter	±5%	SEM
After disk etch	Edge roughness	<1 nm RMS	AFM
After passivation	Stress	<200 MPa	Wafer bow
After metallization	Continuity	100%	Electrical test

50 Reliability Testing

Test	Specification
HTOL	1000h at 125°C
HTSL	1000h at 150°C
Temperature cycling	500 cycles, −40 to 125°C
HAST	96h at 130°C/85% RH
ESD	HBM 2 kV, CDM 500 V
Latchup	100 mA, 85°C
Endurance	10^9 cycles at 85°C
Retention	1000h at 85°C
Read disturb	10^9 reads at 9.886 mV

51 Burn-In

Parameter	Value
Temperature	125°C
Voltage	1.1× nominal
Duration	48 hours
Pattern	Memory exerciser

Part IX
Failure Modes and Mitigations

52 Material Failure Modes

Mode	Cause	Severity	Mitigation
HZO not ferroelectric	Wrong anneal temperature	HIGH	Increase anneal to 500°C
HZO phase degradation	Grain growth at high temperature	MEDIUM	Control RTA profile
Pinholes in HZO	Contamination during ALD	HIGH	Cleanroom, ALD QC
Hf:Zr ratio off	ALD cycle error	HIGH	XPS verification
Film too thick	ALD cycle count error	HIGH	Ellipsometry verification
Film too thin	ALD cycle count error	HIGH	Ellipsometry verification
Moisture absorption	Delay before anneal	HIGH	Anneal within 1 hour
Oxygen vacancy imbalance	Wrong anneal atmosphere	MEDIUM	N_2 anneal
Film thickness variation	ALD non-uniformity	MEDIUM	Ellipsometry at 9 sites
Interface contamination	Poor cleaning	MEDIUM	RCA clean

53 Device Failure Modes

Mode	Cause	Severity	Mitigation
Pillar over-etch	No etch stop	HIGH	Al_2O_3 etch stop layer
Pillar under-etch	Insufficient etch time	MEDIUM	10% overetch
Disk over-etch	Poor selectivity to SiO_2	HIGH	Multi-step etch with endpoint detection
Disk under-etch	Insufficient etch time	MEDIUM	10% overetch
Alignment error	Lithography drift	MEDIUM	± 2 nm overlay tolerance
Edge roughness	Etch non-uniformity	MEDIUM	<1 nm RMS specification
Vortex not forming	Annular width too small	HIGH	Verify annular width is at least $1.5\times$ the vortex core diameter
Vortex drifting	Pinning energy too low	MEDIUM	Verify pillar dimensions

54 Circuit Failure Modes

Mode	Cause	Severity	Mitigation
Comparator offset	Transistor mismatch	MEDIUM	Auto-zeroing
Write driver failure	Voltage wrong	HIGH	Verify write-verify levels
Charge pump failure	Insufficient output	HIGH	Overdesign, verification
ECC failure	Logic error	HIGH	ECC verification
Clock jitter	PLL noise	LOW	<1 ps RMS specification
Power noise	Decoupling insufficient	MEDIUM	On-chip decoupling

55 System Failure Modes

Mode	Cause	Severity	Mitigation
Cell fatigue	Endurance exceeded	MEDIUM	Wear leveling
Weight drift	Retention failure	MEDIUM	Periodic recalibration

Mode	Cause	Severity	Mitigation
Thermal runaway	Cooling failure	HIGH	Thermal shutdown
Read disturb	Read voltage too high	HIGH	Verify read field < coercive field
Package warpage	Stress imbalance	MEDIUM	Stress compensation
ESD damage	Electrostatic discharge	MEDIUM	ESD protection circuits
Latchup	Parasitic SCR	MEDIUM	Guard rings, well taps
Radiation effects	Alpha particles, cosmic rays	LOW	ECC, parity

Part X
Packaging

56 Package Specification

Parameter	Value
Package type	BGA
Ball count	324
Ball pitch	0.8 mm
Package size	19 mm × 19 mm
Package height	1.2 mm
Thermal resistance	15°C/W
Die attach	Epoxy, 25 μm
Wire type	Gold, 25 μm
Bond pad pitch	100 μm

57 Thermal Analysis

Parameter	Value	Notes
Power	97.2 mW	100 mm^2 die
Thermal resistance	15°C/W	Package
Temperature rise	1.5°C	Above ambient
Junction temperature	26.5°C	At 25°C ambient

All operating temperatures remain well within the 0–85°C range. No active cooling is required.

Part XI
Advanced Optimization Available

ADVANCED OPTIMIZATION AVAILABLE

The design specified in this document is the baseline implementation of the ferroelectric vortex memory.
Further optimization is possible.
The advanced optimization provides:

- **Density:** 13.9 GB/cm^2 (7.13× the baseline)
- **Read latency:** 4.02 ns
- **Write energy:** 0.787 fJ per bit (755× lower than the baseline)
- **Power:** 96.9 mW on a 100 mm^2 die
- **Cost per GB:** \$0.405 (6.8× lower than the baseline)

The complete specifications for the advanced optimization are available from the author.
Contact: M.V.Hristov@proton.me
The advanced optimization is not included in this document.

Part XII
Appendix A: Complete Numerical Tables

A Fundamental Engineering Parameters

Parameter	Value
Cell size	80 nm × 80 nm
Disk diameter	40 nm
Pillar diameter	20 nm
Pillar height	1.96 nm
Film thickness	14.83 nm
Annular width	10 nm
Vortex core radius	2.31 nm
Resistance ($n = 1$)	11,762 Ω
Capacitance	23.38 aF
Read voltage	9.886 mV
Read current	840.5 nA
Write voltage (attempt 1)	2.2245 V
Write voltage (attempt 4)	3.1143 V
Erase voltage (attempt 1)	−2.2245 V
Density	1.95 GB/cm^2
Capacity (100 mm^2)	1.95 GB
Read latency	3.80 ns
Write latency (average)	21.1 ns

Parameter	Value
Write energy	595 fJ
Endurance	2.74×10^{10} cycles
Retention at 85°C	>10 years
Power (100 mm^2)	97.2 mW
Die cost	$5.36
Chip cost	$7.36
Cost per GB	$2.75

B Cell Parameters

Parameter	Value
Layer stack (bottom to top)	TiN / Al_2O_3 / TiN pillar / HZO / TiN
Bottom electrode thickness	50 nm
Etch stop thickness	2 nm
Pillar thickness	1.96 nm
Ferroelectric thickness	14.83 nm
Top electrode thickness	50 nm
Cell size	80 nm × 80 nm
Disk diameter	40 nm
Pillar diameter	20 nm
Annular width	10 nm
Select transistor	3-fin FinFET, 28 nm gate
On-current	299 μA
Threshold voltage	0.3 V

C Material Specifications

Parameter	Value
Ferroelectric composition	$Hf_{0.5}Zr_{0.5}O_2$
Ferroelectric thickness	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 $\mu C/cm^2$
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m
Bottom electrode	TiN, 50 nm
Top electrode	TiN, 50 nm
Pillar	TiN, 1.96 nm

Parameter	Value
Etch stop	Al_2O_3, 2 nm

D Process Parameters

Parameter	Value
HZO ALD temperature	250°C
HZO supercycles	82
HZO deposition time	17 min
RTA temperature	450°C
RTA duration	20 s
RTA atmosphere	N_2
Post-etch anneal temperature	400°C
Post-etch anneal duration	30 min
Final anneal temperature	400°C
Final anneal duration	30 min
Cu sheet resistance (1000 nm)	0.017 Ω/□

E Circuit Specifications

Parameter	Value
Comparator decision time	199.5 ps
Comparator power	0.840 μW
Word line driver power	50 μW
Write driver voltage range	±3.2 V
Write driver transistor type	5 V I/O
Charge pump output	−3.2 V
Charge pump efficiency	80%
Charge pump power	302 μW
ECC code type	Hamming SEC
ECC data word	64 bits
ECC code rate	0.90
PLL output frequency	100 MHz to 1 GHz
PLL jitter	<1 ps RMS

F Fabrication Summary

Parameter	Value	Notes	Status
Steps	18	Standard HZO process	Standard
Masks	10		Standard
HZO thickness	14.83 nm	±1%	Specified
Pillar diameter	20 nm	±10%	Specified
Cell size	80 nm	±5%	Specified

Part XIII
Appendix B: Glossary

Term	Definition
Ferroelectric vortex	A topological defect in a ferroelectric nanodisk where the polarization rotates around a central point
Pinning pillar	Central TiN structure that holds the vortex in place
Winding number	Number of times polarization rotates around the pillar
Vortex core radius	Radius of the vortex core region where polarization is suppressed
Annular width	Distance from pillar to disk edge
Annular area	Conductive area of the vortex core annulus
Coercive field	Electric field required to switch polarization
Remanent polarization	Polarization remaining after field removal
HZO	Hafnium zirconium oxide, $Hf_{0.5}Zr_{0.5}O_2$
ALD	Atomic layer deposition
RIE	Reactive ion etching
RTA	Rapid thermal annealing
ECC	Error correction code
BGA	Ball grid array
HTOL	High temperature operating life test
HTSL	High temperature storage life test
HAST	Highly accelerated stress test
ESD	Electrostatic discharge
HBM	Human body model (ESD test)
CDM	Charged device model (ESD test)
Write-verify	Multi-voltage write procedure that maximizes endurance

Part XIV
Appendix C: Licensing

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